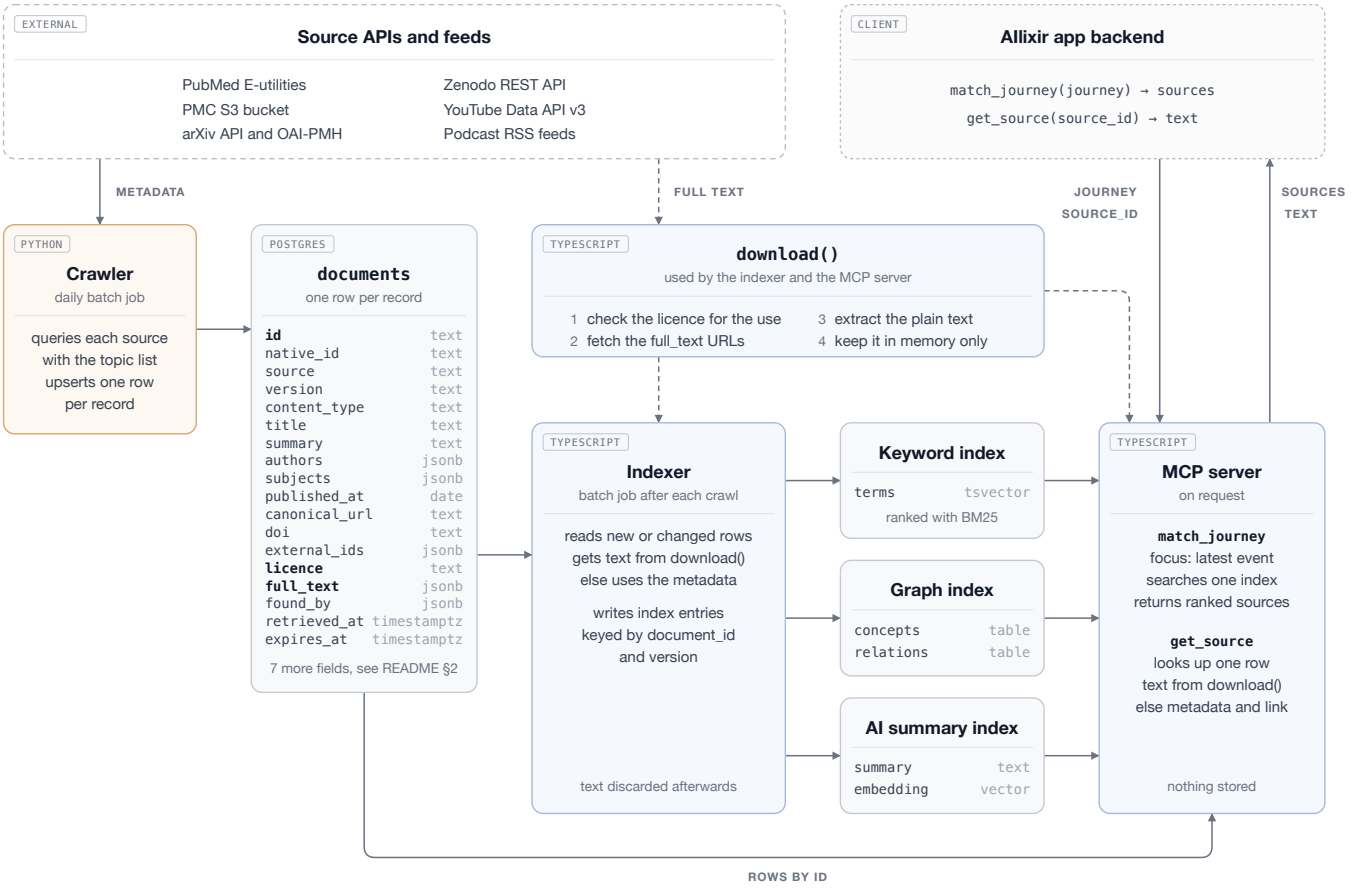
1 Pipeline: crawl, index, match

Reference for the next pages: how a source record moves from the source API to an MCP result. Full texts are never stored.



Python component · TypeScript component · PostgreSQL table · external service · outside our system · data flow · held in memory, not stored

What `documents` stores, so the full source can be fetched later

PURPOSE	FIELDS	USED FOR
Identity	`id` = `<source>:<native_id>`, `source`, `version`	Stable across versions; the MCP `source_id`
Matching	`content_type`, `title`, `authors`, `venue`, `summary`, `subjects`, `language`, `published_at`	All indexes; the only text when the licence blocks the full source
Fetching	`licence`; `full_text` locators in fetch order, each with URL, media type, size and checksum	`download()` checks the licence and fetches without another lookup
Traceability	`canonical_url`, `doi`, `external_ids`, `found_by`, `retrieved_at`	Every result leads back to its source, query and fetch date
Retention	`source_updated_at`, `expires_at`	Re-index on change; YouTube rows refreshed or deleted within 30 days

- **Never stored:** full texts, PDFs, audio, video and transcripts. `download()` holds them in memory only; podcast show notes are cut to 1,000 characters.
- **No full-text cache.** Index rows carry `document_id` and `version` and are rebuilt when the version changes. Feeds are polled with a conditional GET (`ETag`, `Last-Modified`).
- **MCP server:** `match_journey` returns sources from the indexes; `get_source` fetches one source again through `download()`.

2 Sources

What each source provides, checked against the live documentation and with live requests on 01.11.2026.

	PUBMED	PMC	ARXIV	ZENODO	YOUTUBE	PODCASTS
Role	**Primary**, peer-reviewed journals	**Primary**, full texts for PubMed	Secondary, preprints, no review	Secondary, grey literature, no review	Complementary, curated channels	Complementary, curated feeds
METADATA FOR MATCHING						
Title	✓	n/a from PubMed	✓	✓	✓	✓
Abstract, description	✓ abstract	n/a	✓ abstract	✓ description	✓ description, chapters	◑ show notes, cut to 1,000 chars
Authors	✓	n/a	✓	✓ with ORCID	◑ channel	◑ host, rare
Venue	✓ journal	n/a	◑ `journal_ref`	✗	✓ channel	✓ show
Subjects	✓ MeSH	n/a	✓ categories	◑ some MeSH	◑ tags	✗ curated topics
`content_type`	`journal_article`, publication types	n/a	`preprint`	7 resource types	`video`	`podcast_episode`
Published date	✓	n/a	✓	✓	✓	✓
Duration	n/a	n/a	n/a	◑ videos	✓	✓
DOI	◑	◑	◑ journal DOI	✓ own DOI	n/a	n/a cited in notes
Versions	✗	✓	✓	✓ concept ID	n/a	n/a
References	✓ list, `elink`	n/a	◑ Semantic Scholar	◑ sparse	n/a	n/a
FULL SOURCE						
Licence field	✗ see PMC	✓ `license_code`	✓ OAI-PMH, 1 request per paper	✓ `metadata.rights`	◑ `status.license`, not our use	◑ `copyright` only
Full text	✗ via PMCID	✓ JATS XML	✓ HTML, else PDF	✓ files, size, MD5	✗	◑ transcript in 1 of 6 feeds
EXISTING CONNECTORS						
SS26 (`backend/`)	✗ stores PDF text	n/a	✗ stores PDF text	n/a	✗ scrapes captions	n/a
#27 (not merged)	◑ no PMCID, `pdat` filter	n/a	◑ version in ID, no licence	n/a	◑ no `status`, no refresh job	n/a
SIZE, FOR CONTEXT						
Scope	41.4 M citations	about 8 M article versions	about 8,300 a year, medical	3,241 for `prostate AND cancer`	26 curated channels	6 feeds, 715 h

✓ yes ◑ partly ✗ no, or breaks the metadata-only rule n/a not applicable

Full-text route and limits

SOURCE	FULL-TEXT ROUTE	CHECKED BEFORE FETCHING	REQUEST LIMIT	CAP PER QUERY	TERMS THAT AFFECT US
PubMed + PMC	PMC S3 `PMC<n>.<v>.xml` via the PMCID: JATS XML, 60–275 KB, no login	`license_code`	10 requests/s per key, 3 without; 429 above. S3: none published	9,999 IDs	Name NLM; keep data current or mark it as possibly outdated
arXiv	`arxiv.org/html/<id>v<n>`, else `/pdf/`; PDF needs a text extractor	licence	API and OAI-PMH: 1 request per 3 s, one connection. Pages: 15 s crawl delay	30,000	Personal and research use; no serving of e-prints from our servers
Zenodo	File download from the record: PDF, DOCX, PPTX, MP4; up to 50 GB per record	open access, licence, media type, size; MD5 after download	30 searches/min, also with a token; 133 record and file requests/min	10,000	Respect each record's licence; non-military use only
YouTube	None: no transcripts or files. Gemini on the video URL is not cleared	–	10,000 units/day plus 100 `search.list` calls/day	500 per channel search	Refresh or delete stored data within 30 days; no view counts in scores; label as YouTube, never merge
Podcasts	`podcast:transcript` (VTT, SRT, HTML), else show notes. Speech-to-text undecided	copyright	None published; one conditional GET per feed and day	–	No feed grants a licence

Limits apply per key or host and are shared by the crawler, the indexer and the MCP server. Queries above the cap are split by date.

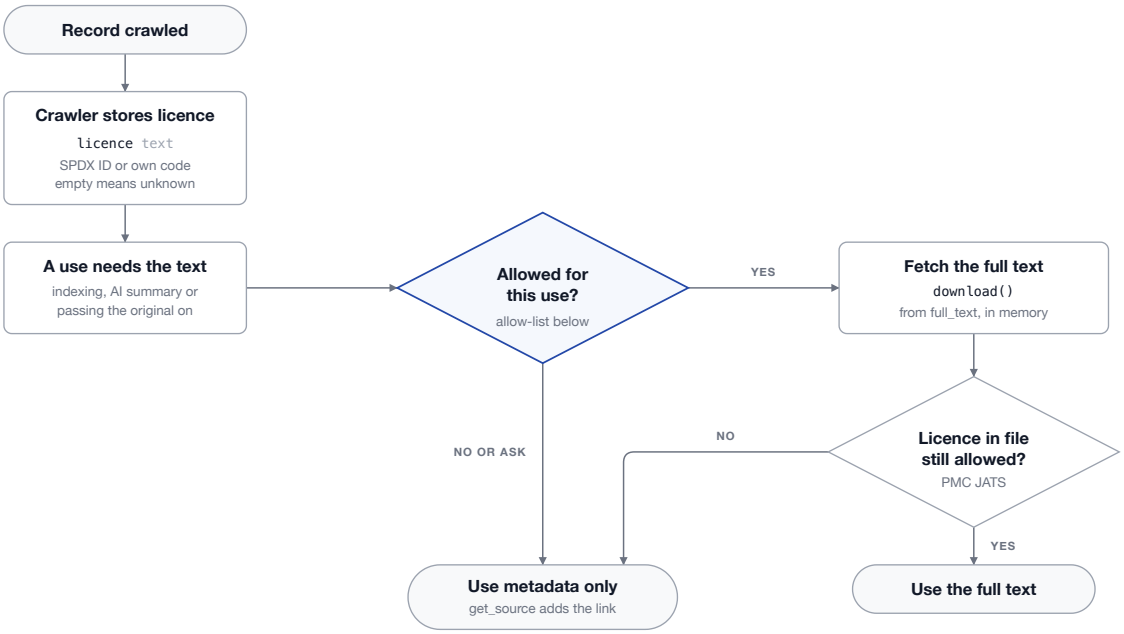
Details, field paths and connector reviews: `Documentation/sources/<source>.md`.

3 Licence handling

The matcher is a non-commercial university project for now. Free to read is not the same as licensed, so every full text passes a licence check before use. `licence` holds an SPDX ID (`CC-BY-4.0`) or our own code (`arxiv-nonexclusive-1.0`, `pmc-tdm`, `youtube-standard`, `all-rights-reserved`).

Why at use time, not in the crawler

- The licence covers the full text, not the metadata (arXiv and Zenodo metadata is CC0, PubMed under NLM terms). Every record stays findable and linkable.
- Most records have no licence when crawled: PubMed without a PMCID, all podcasts.
- Each licence has three permissions, and open or future decisions (commercial use) change them. A new allow-list means re-indexing, not re-crawling. The crawler filters only for relevance.



Allow-list proposal (not legal advice)

LICENCE	KEYWORD, GRAPH, AI INDEX	AI SUMMARY SHOWN TO USERS	ORIGINAL PASSED ON	IF USED COMMERCIALLY LATER
CC0, CC BY, public domain	yes	yes, with credit	yes, with credit	same
CC BY-SA, CC BY-ND	yes	ask	SA: same licence; ND: unchanged only	same
NC licences (CC BY-NC, -NC-SA, -NC-ND)	yes	as the licence without NC	as the licence without NC	no
PMC `TDM`, arXiv default licence	yes (text mining, research use)	ask	no, link only	ask
All rights reserved, unknown	ask	no	no, link only	ask

Where it matters

Licences differ per record, even inside one source: PMC mixes CC BY and NC in one bucket, 8 % of a Zenodo sample and 16 % of an arXiv sample are NC, about a third of medical arXiv papers carry only arXiv's own licence, and no podcast feed grants one. YouTube's 30-day rule applies whatever `status.license` says.